



Submersible Grey Water Pump

AmaDrainer 301/302/303/358

60 Hz

Type Series Booklet



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Type Series Booklet AmaDrainer 301/302/303/358

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Drainage

Drainage Pumps / Grey Water Pumps

AmaDrainer 301/302/303/358



 Size 358 not available in 60 Hz version

Main applications

- Drainage
- Disposal
- Drainage systems
- Lowering groundwater levels
- Maintaining groundwater levels
- Draining of pits, shafts, etc.
- Surface drainage

Fluids handled

- Solid particles with a particle size of up to 10 mm
- Slightly contaminated grey water ($\leq 50^\circ\text{C}$)
- Wash water (up to $\leq 90^\circ\text{C}$ max. for short periods $t \leq 3$ minutes)
- Seepage water
- River water, lake water and groundwater
- Brackish water
- Seawater ($\leq 20^\circ\text{C}$)
- Water containing salt ($\leq 20^\circ\text{C}$)

Further information on fluids handled

Overview of fluids handled ($\Rightarrow$ Page 6)

Operating data

Table 1: Operating properties

Characteristic		Value
Flow rate	Q [m^3/h]	$\leq 16,5$
	Q [l/s]	$\leq 4,6$
Head	H [m]	$\leq 12,5$
Fluid temperature	T [$^\circ\text{C}$]	$\leq +50$ (continuous duty)
		$\leq +90$ (short-time duty)

Design details

Design

- Fully floodable submersible motor pump
- Close-coupled design
- Single-stage
- Wetted parts made of materials coated with anti-corrosive
- External float switch
- Maximum immersion depth: 2 m
- Integrated swing check valve

Installation

- Vertical installation
- Wet-installed transportable model

Drive

- Jacket-cooled single-phase AC motor
- 220 V / 240 V, 60 Hz
- Frequency of starts ≤ 30 starts/hour
- Integrated temperature switch
- Enclosure IP68 (permanently submerged) to EN 60529/ IEC 529
- Power cable incl. shockproof plug

Shaft seal

- Drive end, 1 shaft seal ring
- Pump end, 2 shaft seal rings
- Liquid reservoir between the seals for cooling and lubrication

Bearings

- Rolling element bearings
- Grease-packed for life
- Maintenance-free

Automation

- Pump controlled via float switch
- Pump controlled by external control unit

Designation

Example: AmaDrainer C 301

Table 2: Designation key

Code	Description
AmaDrainer	Type series
C	Material variant
	A ¹⁾ Variant for grey water
	C ²⁾ Universal variant, also for aggressive fluids
301	Size

Materials

Table 3: Overview of available materials

Part No. (⇒ Page 14)	Description	Material
79-1	Float switch (float)	Polypropylene
101	Pump casing	Polypropylene, 20 % glass fibre reinforced
107	Discharge casing	Polypropylene, 30 % glass fibre reinforced
161	Casing cover	Polypropylene with 20 % talc (PP-TS 20)
210	Shaft	Chrome nickel molybdenum steel (1.4404)
230	Impeller	Noryl GNF2
811	Motor housing	Chrome nickel molybdenum steel (1.4404)

Product benefits

- **Universal use, provides peace of mind regarding the fluids handled**
 - Supplied in material variant C as standard
 - **Absolutely tight**
 - Hermetically sealed pump head, automatic switchgear activated by a magnetic switch
 - **Flexible adjustment**
 - Automatic switchgear with adjustable, variable level control and maximum travel stop
 - Straightforward adaptation for external control system by means of supplied locking disc
 - **Easy and fast replacement**
 - Power cable with mains plug
 - Discharge-side connection via connection socket
 - **No drainage of the discharge pipe**
 - Integrated swing check valve
 - **Minimal space requirements**
 - Compact design with vertical discharge nozzle for installation in narrow sumps
 - **Protected against overload**
 - Integrated thermal motor protection
 - **High operating reliability**
 - Motor protected by triple shaft seal lubricated by the barrier fluid
 - Additional lubrication of the seal faces in the event of dry running
 - **Durable and robust**
 - High-quality plastic casing provides low weight, high corrosion resistance, good noise absorption, abrasion resistance and impact resistance
 - **Economical operation**
 - Hydraulic system and motor optimised for high pump output at high efficiency
- **Protected against overheating**
 - The jacket-cooled motor design permits operation even when the motor is not submerged.

1 Size 358 only
2 Sizes 301/302/303 only

Overview of product features / selection tables

Overview of fluids handled

Table 4: Symbols key

Symbol	Description
X	Standard
-	Version not available / not feasible

The table of fluids handled is a selection aid for different applications. It serves as guidance and is based on long-standing experience. The data are reference values and are not to be considered generally binding recommendations. They shall not be the basis for warranty claims. Please contact your nearest sales branch for in-depth advice.

Table 5: Table of fluids handled

Fluids handled	Material variant C
Antifrogen-water mixture	X
Cider	X
Beer	X
Spirits	X
Buttermilk	X
Calcium acetate	X
Calcium hydroxide	X
Deionised water	X
Peanut oil	X
Vinegar	X
Ethylene glycol	X
Liquid fertiliser	X
Anti-freeze	X
Glycerine	X
Grisiron	X
Uric acid	X
Potassium hydroxide	X
Calcium carbonate	X
Coconut oil	X
Evaporated milk	X
Lemonades	X
Corn oil	X
Milk	X
Whey	X
Sodium carbonate	X
Sodium chloride up to 15 % concentration	X ⁽³⁾
Sodium hydrogen phosphate	X
Sodium nitrate	X
Sodium perborate	X
Sodium sulphate	X
Polyglycols	X
Rapeseed oil	X
Castor oil	X
Silicone oil	X
Silo leachate	X
Soybean oil	X
Edible vinegar	X
Edible oil	X
Trisodium phosphate	X
Vaseline	X
Washing machine lye	X
Water	
Drainage water	X

3 Fluid temperature ≤ 25 °C

Fluids handled	Material variant C
Fire-fighting water	X
Heating water	X
Lime water	X
Boiler water	X
Condensate	X
Cooling water	X
Seawater	X ⁴⁾
Raw water	X
Swimming pool water (DIN 19643)	X
Salt water	X ⁴⁾
Partly desalinated water	X
Fully desalinated water	X

Technical data

AmaDrainer 301/302/303/358

Table 6: Technical data, 220 V / 240 V, 60 Hz

Size	Discharge-side connection	Free passage	P ₁	P _N	I _N 1~230 V	Power cable	Mat. No.	[kg]
						H05 RN8		
		[mm]	[kW]	[kW]	[A]	[m]		
Material variant C (variant for grey water)								
C 301	Rp 1 1/4	10	0,50	0,21	2,15	10	48034492	5.2
C 302	Rp 1 1/4	10	0,75	0,36	3,30	10	48034493	6.7
C 303	Rp 1 1/4	10	1,20	0,60	5,00	10	48034494	6.9

Table 7: Technical data, 220 V / 240 V, 60 Hz (for the UK)

Size	Discharge-side connection	Free passage	P ₁	P _N	I _N 1~230 V	Power cable	Mat. No.	[kg]
						H05 RN8		
		[mm]	[kW]	[kW]	[A]	[m]		
Material variant C (universal variant, also for aggressive fluids)								
C 301	Rp 1 1/4	10	0,50	0,21	2,15	10	48034495	5.2
C 302	Rp 1 1/4	10	0,75	0,36	3,30	10	48034496	6.7
C 303	Rp 1 1/4	10	1,20	0,60	5,00	10	48034497	6.9

Characteristic curves

AmaDrainer 301/302/303; n = 3500 rpm (multi-vane impeller)

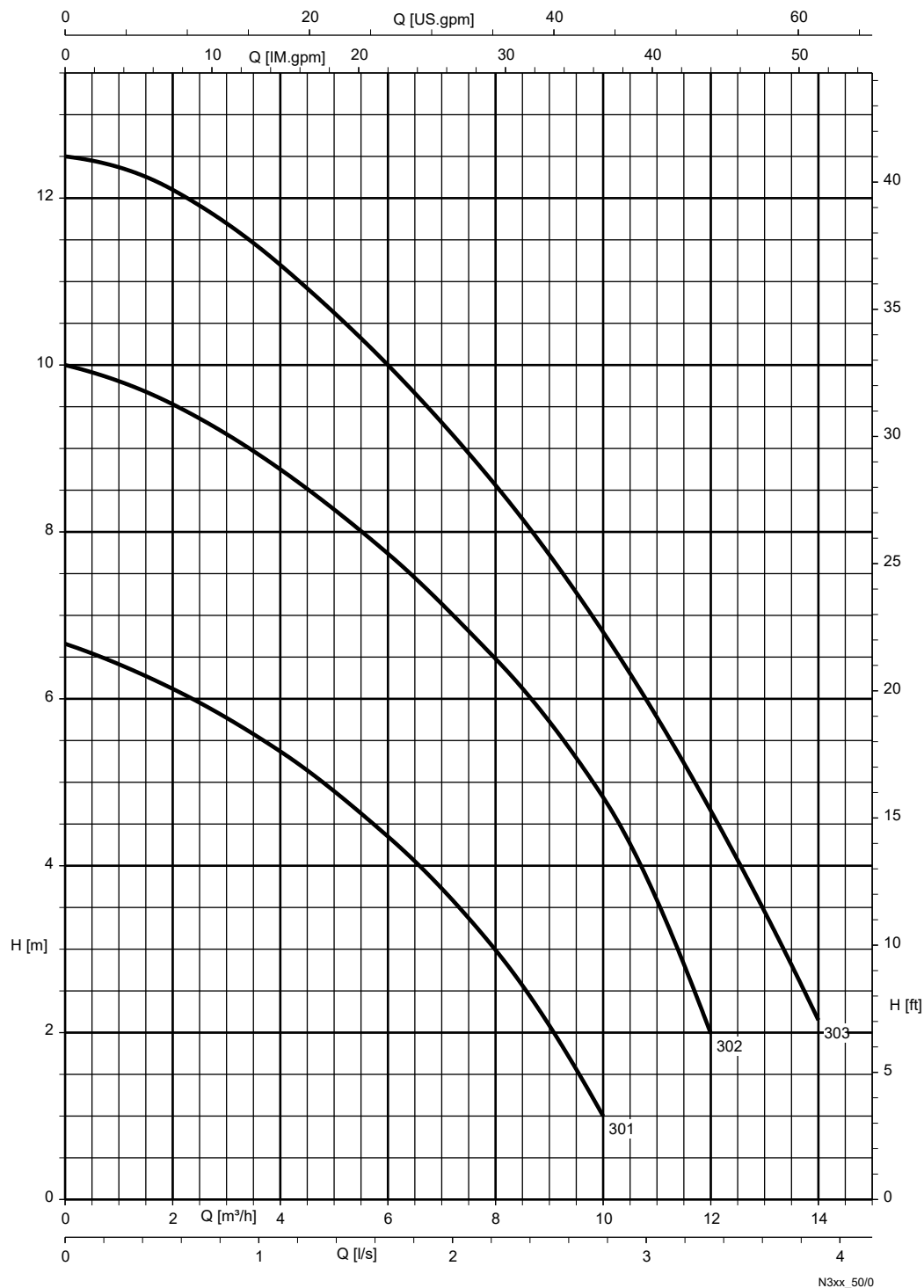


Fig. 1: Sizes 301/302/303: free passage = 10 mm

Performance tolerance to ISO 9906, Annex A (water under standard conditions)

Dimensions and connections

AmaDrainer 301/302/303/358

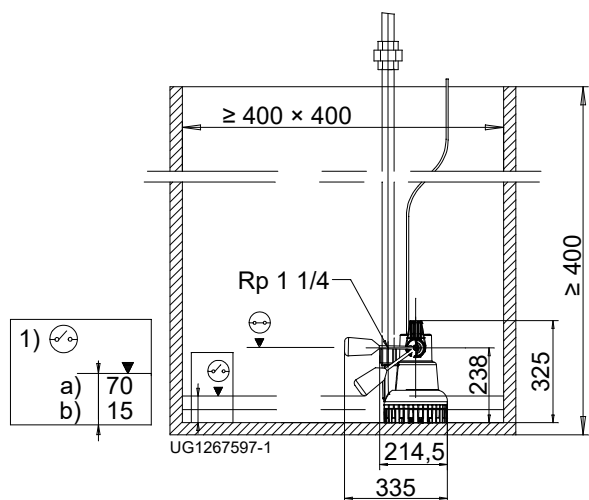


Fig. 2: Dimensions of AmaDrainer 301

1)	Residual water level
a)	Automatic
b)	Manual

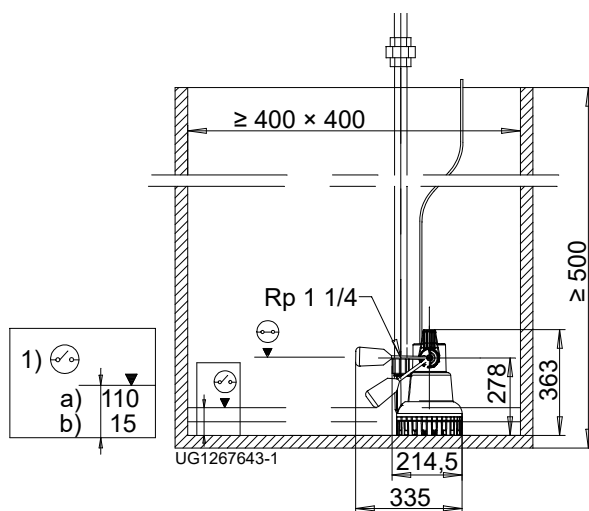


Fig. 3: Dimensions of AmaDrainer 302/303

1)	Residual water level
a)	Automatic
b)	Manual

Installation information

Installation example of a dual-pump station

i Dual-pump stations are operated via an external control system. For this purpose, the float switch suspension must be locked in place using the locking disc included in the scope of supply.

i The inlet is positioned above the highest float switch.

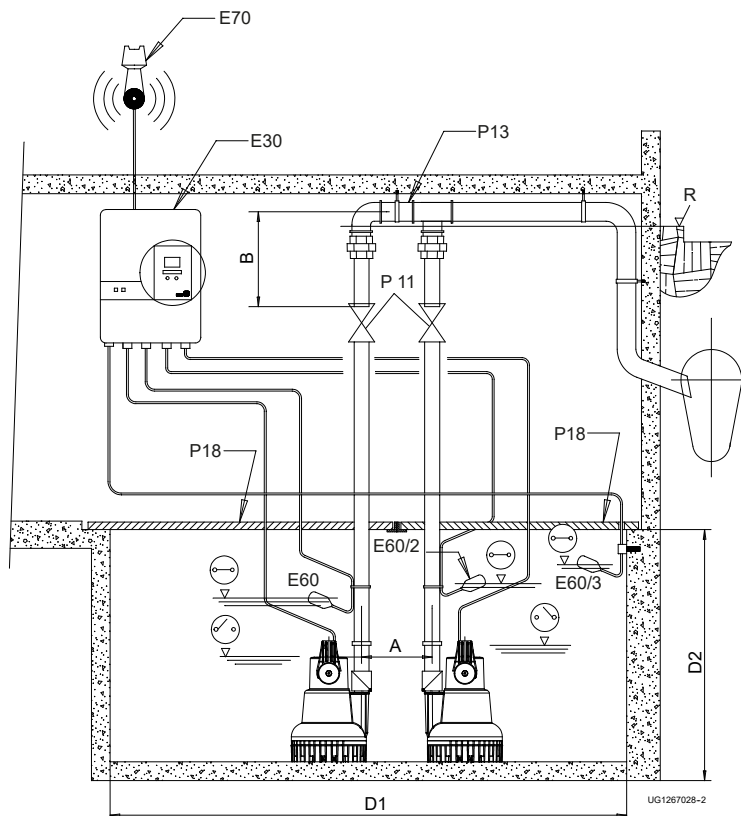


Fig. 4: Installation example of a dual-pump station

1)	Residual water level
a)	Automatic
b)	Manual
E30	Control unit
E60	Float switch, base load
E60/2	Float switch, peak load
E60/3	Float switch, high water alert
E70	Horn
P11	Gate valves
P13	Y-pipe
P18	Cover plate
R	Flood level

Table 8: Dimensions of a dual-pump station

Size	A	B	D1	D2
	[mm]	[mm]	[mm]	[mm]
301	275	320	1060 × 500	400
302	275	320	1060 × 500	500
303	275	320	1060 × 500	500

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set
- Installation/operating manual
- Locking disc for continuous operation
- Integrated swing check valve
- Connection socket with internal thread

Accessories

Pump accessories

Table 9: Pump accessories

Item	Description	Connection	AmaDrainer			Mat. No.	[kg]
			301	302	303		
P5 	Discharge-side adapter, graded For connecting various discharge lines (connections Rp 1 1/4 to Rp 3/4, DN 25 or DN 32)	Rp 1 1/4 to Rp 3/4, DN 25 or DN 32	X	X	X	05086820	0.16
P10 	RK swing check valve Plastic, EN 12050-4, with internal thread ISO 7/1, with full bore and drain plug	Rp 1 1/4	X	X	X	01009771	0.1
P10 	Lift check valve, stainless steel (1.4401)	Rp 1 1/4	X	X	X	01084936	2.1
P11 	Socket gate valve Material: CuZn, PN 16, with internal thread, with full bore	Rp 1 1/4	X	X	X	01014219	0.63
P13 	Y-pipe For dual-pump sets, with external thread, galvanised steel	Rp 1 1/4	X	X	X	18040311	4.1
P18 	Cover plate Tread-proof, split, with profile seals and angle iron mounting frame (type A 560) for 500 x 500 mm pits (For dual-pump stations with Y-pipe 2 cover plates are fitted next to each other.)	Rp 1 1/4	X	X	X	18075627	13
P21 	A 25 B drain hose set (see P32) Comprising: rigid coupling with external thread, 6 m plastic hose DN 25, rapid-action hose coupling Rp 1 1/4, (free passage 21 mm)	Rp 1 1/4	X	X	X	18079719	3
P24 	Storz rigid coupling with internal thread to DIN ISO 228/1 Aluminium alloy, required pipe part see P32 Can be combined with P28	C-G 1 1/2	X	X	X	01002463	0.3
P26 	Storz hose coupling Required counterpart see P24	C 52 (DIN 14321)	X	X	X	00524551	0.3

Item	Description	Connection	AmaDrainer			Mat. No.	[kg]
			301	302	303		
 P28 Synthetic hose DN 40, with integrated C couplings, DIN 14811 Required pipe part see P32 Can be combined with P24		C 42-5 m	X	X	X	01062592	1.7
		C 42-10 m	X	X	X	01062593	2.8
		C 42-20 m	X	X	X	01062594	5
	Synthetic hose DN 50, DIN 14811, with integrated C couplings Required pipe part see P32 Can be combined with P24	C 52-5 m	X	X	X	00522262	2.3
		C 52-10 m	X	X	X	00522263	4.2
		C 52-20 m	X	X	X	00522264	5.7
 P31 Bellmouth for draining residual water		-	X	X	X	39300101	0.2
 P32 Pipe extension for Storz rigid coupling C (P24) PVC hard, internal/external thread For use without swing check valve and union nut		Internal thread Rp 1 1/4 / external thread R 1 1/2 x 170	X	X	X	11035587	0.2

Control units and switchgear

 Control units and switchgear on request

Exploded view with list of components

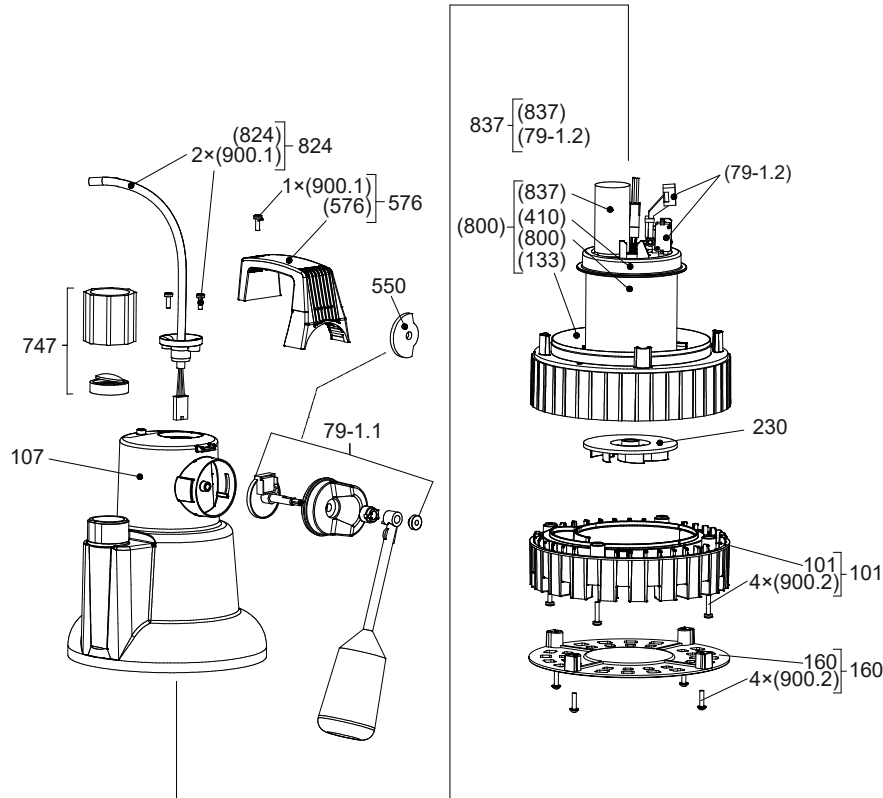


Fig. 5: Exploded view

Table 10: List of components

Part No.	Description	Part No.	Description
101	Pump casing	747	Swing check valve
107	Discharge casing	79-1.1	Automatic switchgear, external
133	Clearance plate	79-1.2	Automatic switchgear, internal
160	Cover	800	Motor
230	Impeller	824	Power cable
410	Profile seal	837	Capacitor
550	Locking disc for continuous operation	900.1/.2	Screw
576	Handle		



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